The examination is being carried out on the **following application documents**

Description, Pages

1-65 as published

Claims, Numbers

1-15 filed in electronic form on 10-10-2024

Drawings, Sheets

1/27-27/27 as published

CITED PRIOR ART

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 US 2021/281595 A1 (HIRANO RYO [JP] ET AL) 9 September 2021 (2021-09-09)

D2 US 2017/201537 A1 (CALDWELL JEFFREY [US] ET AL) 13 July 2017 (2017-07-13)

1 The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claim 11 is not new within the meaning of Article 54(1) and (2) EPC.

1.1 Document D1 discloses the following features of claim 1 (remarks in parentheses refer to D1):

A method (**[2]**: "The present disclosure relates to an anomaly detection device and an anomaly detection method for detecting anomalous communication in an in-vehicle network."), **comprising:**

interpreting, via a network monitoring component of a controller, network communications associated with at least one of a plurality of network zones of a vehicle (**[16]**: "a detection rule generator that monitors a communication establishment frame flowing over the Ethernet in a communication establishment phase of the service-oriented communication and that generates, for each communication ID"), **each network zone of the plurality comprising a**

plurality of end points (**[58]**: " Central ECU 200 controls Zone ECUs 300a, 300b, 300c, and 300d and controls in-vehicle network system 10 as a whole."; **[60] to [63]:** "Zone ECU 300a communicates with Central ECU 200, IDS ECU 100, and Zone ECUs 300b, 300c, and 300d via Ethernet 13 and communicates with camera ECU 400a via Ethernet 12 to control ON/OFF of a camera image. Zone ECU 300b communicates with Central ECU 200, IDS ECU 100, and Zone ECUs 300a, 300c, and 300d via Ethernet 13 via Ethernet 11 to control the display of the car navigation system. Zone ECU 300c communicates with Central ECU 200, IDS ECU 100, and Zone ECUs 300a, 300b, and 300d via Ethernet 13 and communicates with steering ECU 400c via CAN 14 to control the steering of the steering wheel. Zone ECU 300d communicates with Central ECU 200, IDS ECU 100, and Zone ECUs 300a, 300b, and 300c via Ethernet 13 and communicates with brake ECU 400d via CAN-FD 15 to control the brake.");

detecting, via a network intrusion detection component of the controller, an intrusion event in response to the network communications (**[16]**: "an anomaly detector that monitors a communication frame flowing over the Ethernet in a communication phase of the service-oriented communication and that, by referring to a detection rule that includes a communication ID written in the communication frame, detects the communication frame as an anomalous frame when a server address or a client address written in the communication frame differs from a server address or a client address included in the detection rule"); **and**

performing, via an intrusion response component of the controller, an intrusion response operation in response to the detected intrusion event (**[16]**: "an anomaly notifier that provides a notification of an anomaly in response to the anomalous frame being detected").

1.2 Therefore, the subject-matter of independent claim 11 is not new (Article 54(1) and (2) EPC).

2 The subject-matter of **independent claim 1** is already covered in the analysis of claim 11 and therefore also considered not new.

3 The **dependent claims** do not contain any features which, in combination with the features of any claim to which they refer, meet the requirements of the EPC in respect of novelty or inventive step, in particular:

3.1.1 Claims 2, 12: See the citation above for claim 11.

3.1.2 Claims 3, 13: See **[62] in D1**: "CAN 14".

3.1.3 Claims 4, 14: See **[159] in D1**: "if the current vehicle state acquired from vehicle state extractor 170 fails to match the vehicle state in the aforementioned row".

3.1.4 Claims 5, 15: Suggested by **D2, [43]**: "frequency".

3.1.5 Claim 6, 7: Suggested by **D2, [101]**: "payload"; "destination field".

3.1.6 Claim 8: See the citation above for claim 11.

3.1.7 Claim 9: Issuing a notification based on the severity of a detected intrusion event is a normal design option.

3.1.8 Claim 10: See [141] in D1: "anomaly notifier 160 transmits, to Central ECU 200, a frame requesting that the driver or an emergency report destination, another vehicle, another system, or another Intrusion Prevension System (IPS) be notified of the anomaly in the vehicle".

4 **Formal remarks**

4.1 To meet the requirements of Rule 42(1)(b) EPC, documents D1 and D2 should be identified in the description and the relevant background art disclosed therein should be briefly discussed.

4.2 The features of the claims should be provided with reference signs placed in parentheses to increase the intelligibility of the claims (Rule 43(7) EPC).